Appendix

GIV in Practice: PCA and GMM Implementation

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The Need for Γ: Leaky Instruments

If large firms respond differently to macro trends, the simple GIV fails.

- **The Problem:** If big firms are more sensitive to macro shocks (η_t), then $y_{St} - y_{Et}$ is correlated with the error term.
- **The Solution:** We need weights Γ such that $\sum \Gamma_i = 0$ AND $\sum \Gamma_i \lambda_i = 0$.
- **The Formula:** $\Gamma = S - \lambda(\lambda'\lambda)^{-1}\lambda' S$ (the residual of S on λ).

Implementation 1: PCA (The Two-Step Approach)

Simplest when you have a long panel and common factors dominate.

Algorithm:

1. **De-mean:** Center the outcome matrix Y (firm-by-time).
2. **Factorize:** Run PCA on Y and extract the first few loadings $\hat{\lambda}_i$.
3. **Project:** Calculate Γ_i as the residuals of regressing size S_i on $\hat{\lambda}_i$.
4. **Instrument:** Construct $z_t = \sum \Gamma_i y_{it}$ and use in standard 2SLS.

Pro: Computationally cheap. Con: Ignores the endogenous variable's impact on factors.

Implementation 2: Iterative GMM (The Robust Approach)

Simultaneously identifies the elasticity (ψ) and the loadings (λ_i).

Iterative Algorithm:

- **Start:** Assume $\psi^{(0)} = 0$ and $\lambda_i^{(0)} = 1$.
- **Loop:**
  1. **Residuals:** Calculate $e_{it} = y_{it} - \psi^{(k)} x_{it}$.
  2. **Loadings:** Extract $\lambda_i^{(k+1)}$ from e_{it} (e.g., via PCA on residuals).
  3. **GIV:** Update $z_t = \sum \Gamma(\lambda^{(k+1)}) y_{it}$.
  4. **Estimate:** Update $\psi^{(k+1)}$ via IV regression of y on x using z_t.
- **Converge:** Stop when ψ and λ stabilize.

Summary: When to use which?

Method	Complexity	Ideal Use Case
Simple GIV	Low	Homogeneous firms ($\lambda_i \approx \bar{\lambda}$)
PCA GIV	Medium	High Signal-to-Noise, T is large
GMM GIV	High	Endogeneity in factor structure

Bottom Line: In granular economies, **always** check if results are robust to using Γ (the "Purified" GIV weights).